

# Devin Johnson

## Senior AI Engineer

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### Summary:

Senior AI Engineer with over 10 years of experience designing and deploying large-scale, real-time **machine learning** and **deep learning** systems in streaming media and fintech sectors. Skilled in AI-driven **recommendation systems**, **NLP**, **MLOps**, and distributed model serving on **Kubernetes**. Delivered enterprise-scale solutions enhancing engagement, reducing latency, and boosting operational reliability by 30%. Proven leadership in cross-functional project direction and production AI **pipeline optimization**.

### Experience:

#### Senior AI Engineer | Roku | 09/2023 - Present

Elevated Roku's **machine learning** capabilities in **recommendation systems**, enhancing user engagement and ad revenue significantly.

- Architected **deep learning** models lifting engagement 18% across 70M+ accounts using PyTorch and **AWS**.
- Developed real-time personalization models, boosting click-through rates by 22% and ad revenue 19% in two quarters.
- Engineered low-latency pipelines with ONNX, reducing serving latency by 35% over 5B+ daily predictions.
- Built feature pipelines cutting model training cycle times by 40% and eliminated training/serving skew.
- Designed an A/B/n framework increasing retention by 12% and tripling decision velocity.
- Implemented **model deployment** strategies reducing deployment time from days to hours.
- Scaled model serving on **Kubernetes** maintaining 99.99% availability for 5B+ predictions.

#### Senior Data Engineer | Mycroft AI Inc | 05/2019 - 02/2023

Led the development of a robust data infrastructure supporting AI advancements in voice recognition technologies.

- Architected data platforms supporting Transformer-based ASR model training for voice AI products at petabyte scale.

- Optimized feature pipelines processing 10M+ sessions daily, enhancing efficiency by 45%.
- Designed lakehouse architectures on BigQuery, cutting data preparation time by 80%.
- Implemented real-time data streaming with **Kafka** and **Flink** for rapid inference feedback loops.
- Migrated cloud infrastructure from **AWS** to **GCP**, cutting costs by 20% (~\$480K annually).
- Sustained data reliability at 99.9% by establishing robust pipeline monitoring and alerting systems.

## **Software Engineer | Enova International | 08/2016 - 04/2019**

Enhanced system efficiency and integrated ML models to improve financial decision-making processes.

- Integrated ML models for credit scoring, reducing loss rates by 18% through more accurate predictions.
- Developed backend services using **Python** and **Go** supporting 5K+ requests/sec with <100ms latency.
- Optimized service architecture, reducing response times by 30% via model serialization and caching.
- Deployed models using shadow traffic and automated rollbacks, improving deployment reliability.
- Refactored monolithic systems into **microservices**, facilitating better ML integration and scalability.

## **Data Science Researcher | Civis Analytics | 06/2014 - 08/2016**

Provided advanced predictive analytics to enhance audience targeting and engagement strategies.

- Built predictive models boosting audience engagement using data from millions of records.
- Applied uplift modeling in **A/B testing**, quantifying the impact of campaigns for enterprise clients.
- Engineered features from diverse datasets, improving model accuracy by 12-18%.
- Developed and maintained **Python** and SQL pipelines enhancing data processing and **model evaluation**.

## **Skills:**

- **Frontend:** React, Next.js, TypeScript
- **Backend:** Python, Go, Microservices, REST, gRPC, Event-Driven Architecture
- **Databases:** PostgreSQL, MongoDB, Redis, Elasticsearch
- **Cloud & DevOps:** AWS, GCP, Kubernetes, Docker, Terraform, Jenkins, **CI/CD**, Prometheus
- **Testing & Automation:** Model Evaluation, A/B Testing, Automated Rollouts

- **AI/ML & Data:** Machine Learning, Deep Learning, Transformers, BERT, Recommendation Systems, **NLP**, **Model Monitoring**, Drift Detection, **Vector Databases**
- **Other Tools:** Apache Spark, Airflow, Kafka, Flink, Triton Inference Server

## **Education:**

### **B.S. in Computer Science**

Westwood College

Graduated May 2014